

PARISIAN RUIN OF BRANCHING BROWNIAN MOTION

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ABSTRACT. We study infinite-horizon Parisian simultaneous ruin of a branching Brownian population with prescribed offspring numbers $2 \leq k_i \leq K < \infty$ at successive splits. All particles currently alive must exceed a linear boundary throughout one common window, or at every observation in a consecutive grid window. Sibling Brownian motions may have arbitrary prescribed correlation matrices, including singular matrices. For every fixed window we obtain an exact exponential asymptotic whose constant is a sliding-window Gaussian functional averaged over the genealogy and, for discrete observation, over the grid phase. The window start and completion have the same conditional normal limit. We also establish the zero-window and dense-grid limits of the constants. Ordinary simultaneous ruin is included by taking a window of length zero; in that case an exact integral identity gives an exponentially smaller remainder.

1. INTRODUCTION AND MAIN RESULT

We consider a Brownian population started by one particle at the origin. Each particle has an independent exponential splitting clock of rate $\beta > 0$. At the i th split in chronological order, the parent is replaced by k_i children, where $2 \leq k_i \leq K < \infty$ is a prescribed sequence. Every ancestral path is marginally a standard Brownian motion. Sibling motions may be correlated through a prescribed matrix A_i , while different birth events use independent Gaussian drivers. Both the offspring numbers and the sibling covariance matrices may vary between splits.

Against the linear boundary $u + ct$, with $c > 0$, we study two infinite-horizon Parisian ruin probabilities. For *branch ruin*, one ancestral line must remain above the boundary throughout an interval of length h , continuing through descendants if splits occur within the interval. For *simultaneous ruin*, every particle alive at each time of one common interval must be above the boundary. A window of length zero gives the corresponding classical ruin event. Our main theorem treats simultaneous ruin under continuous and uniform discrete observation in the model above. We also study continuous Parisian branch ruin for homogeneous offspring with independent sibling motions.

The closest branching results concern classical ruin. For binary branching Brownian motion, exact infinite-horizon branch-ruin asymptotics are obtained in [7, Proposition 4], formulated there through the all-time minimum and related to our event by reflection. Classical branch and simultaneous ruin over a finite horizon are studied in [6] for Brownian decision trees with prescribed branching times and possibly different offspring numbers. Finite-horizon simultaneous-ruin asymptotics for classical branching Brownian motion with exponential branching times are also given in [6, Proposition 1.1]. Here we consider persistence over a fixed Parisian window and an infinite observation horizon.

For nonbranching models, Parisian ruin has been studied for Gaussian risk processes in [2, 3], for scalar Brownian models under discrete observation in [4], and for two correlated Brownian reserves in [5]. An exact infinite-horizon formula for spectrally negative Lévy processes with a fixed Parisian delay, including Brownian motion with positive drift, is given in [1].

For simultaneous ruin we obtain an equivalent $Ce^{-\kappa u}$, where $\kappa = c + \sqrt{c^2 + 2\beta}$. The offspring and covariance sequences, Parisian window and observation mesh enter the constant C , while the exponential rate depends only on c and β . The constant is an explicit sliding-window Gaussian functional averaged over the genealogy and, for discrete observation, over the observation phase. Conditional on ruin, the window start and completion have the same normal limit, with center u/a and scale $\sqrt{u/a^3}$, where $a = \sqrt{c^2 + 2\beta}$. We also establish the zero-window and dense-grid limits of the constants.

For continuous branch ruin with $k_i \equiv k$ and $A_i = I_k$, we obtain an equivalent $B_{k,h}e^{-\lambda u}$ when $c > \sqrt{2\beta(k-1)}$, where $\lambda = c + \sqrt{c^2 - 2\beta(k-1)}$. Its Pickands–Berman representation uses one genealogical cluster for every delay: only the window functional changes, and shrinking the window recovers the classical constant. The varying-offspring scope differs between the two ruin events. A bounded prescribed offspring sequence can prevent even a limiting logarithmic rate for branch ruin; we give a counterexample with binary and ternary splits.

Our simultaneous-ruin proof conditions on the first split and its Brownian endpoint. A direct Brownian-bridge coupling gives convergence of the full sliding-window functional, while a weighted binary subtree supplies the uniform exponential moment needed for the infinite horizon. For homogeneous branch ruin, the first-hitting-line identity shows that every fixed Parisian delay is an exact spatial shift of the classical ruin curve. A separate occupation-normalized spine argument identifies the Pickands–Berman constant.

Section 2 presents the ruin-time limit, properties of the simultaneous constants, the exact ordinary-ruin identity, random environments, homogeneous branch ruin and the obstruction to arbitrary varying offspring. Sections 3 and 4 give the respective proofs. Appendix A contains the uniform estimates and the varying-offspring counterexample; the spectral interpretations are collected in Appendix B.

Fix a boundary slope $c > 0$, a splitting rate $\beta > 0$, and a deterministic sequence

$$(1.1) \quad \begin{aligned} \mathcal{E} &= ((k_i, A_i))_{i \geq 1}, \quad 2 \leq k_i \leq K < \infty, \\ A_i &\in \mathbb{R}^{k_i \times k_i}, \quad R_i = A_i A_i^\top, \quad (R_i)_{jj} = 1 \quad (1 \leq j \leq k_i). \end{aligned}$$

Each k_i is an integer. The sequence and its bound K do not depend on u . No convergence, periodicity or nonsingularity is assumed. The off-diagonal entries of R_i may have mixed signs. The normalization makes every individual Brownian motion standard.

Start with one particle at the origin, moving as a standard Brownian motion B_0 . Every particle has an independent exponential lifetime of rate β , independent of all Gaussian drivers. At the i th split in chronological order, the parent is replaced at its current position by k_i children. Assign this birth event a fresh vector

$$\mathbf{W}^i(t) = A_i \mathbf{B}^i(t), \quad t \geq 0,$$

where the k_i coordinates of $\mathbf{B}^i$ are independent standard Brownian motions. The vectors for distinct events are independent of one another and of B_0 . A child uses its coordinate until its own split. Its descendants then use a new vector, while its surviving siblings retain their original coordinates. Thus

$$\text{Cov}(W_j^i(s), W_l^i(t)) = (R_i)_{jl} \min(s, t).$$

Births are simultaneous within a cohort; the independent lifetime clocks almost surely have no ties between splitting events. Taking $k_i \equiv k$ and $R_i \equiv I_k$ gives the usual k -ary branching Brownian motion.

After j splits the population has $n_j = 1 + \sum_{i=1}^j (k_i - 1) \leq 1 + j(K - 1)$ particles, with $n_0 = 1$. The successive holding rates are βn_j . Comparison with the pure birth process having rates

$\beta(1 + j(K - 1))$ proves nonexplosion. In particular, only finitely many births occur on every compact time interval. Let $N(t)$ be the living population and $X_v(t)$ the position of $v \in N(t)$. Since children start at the parent's position and there is no extinction, the lower envelope

$$(1.2) \quad Y(t) = \min_{v \in N(t)} X_v(t) - ct, \quad t \geq 0,$$

is continuous. For a terminal time T , the minimum of the ancestral paths of the particles in $N(T)$ agrees with the minimum over $N(t)$ at every $t \leq T$.

The same conclusions hold when the prescribed pairs (k_i, A_i) are indexed by generation, with the ancestor using the first pair at its split. Both constructions have a finite genealogy on compact intervals and a post-first-split cohort law independent of the absolute split time. We use chronological indexing below.

For $h \geq 0$ define the first successful window start and its completion by

$$(1.3) \quad S_u^{0,h} = \inf \left\{ t \geq 0 : \inf_{0 \leq r \leq h} Y(t+r) > u \right\}, \quad T_u^{0,h} = S_u^{0,h} + h.$$

We use $\inf \emptyset = \infty$ and put $\mathcal{A}_{0,h}(u) = \{S_u^{0,h} < \infty\}$ and $p_{0,h}(u) = \mathbb{P}(\mathcal{A}_{0,h}(u))$. Equivalently, there is a common interval of length h on which every reserve $u + ct - X_v(t)$ currently present is negative. Crossing does not remove a particle. The start $S_u^{0,h}$ is defined using the whole future window; $T_u^{0,h}$ is its completion time.

For a mesh $\delta > 0$ and $m \in \mathbb{N}_0$, define

$$(1.4) \quad S_u^{\delta,m} = \delta \inf \left\{ j \in \mathbb{N}_0 : \min_{\ell=0,\dots,m} Y((j+\ell)\delta) > u \right\}, \quad T_u^{\delta,m} = S_u^{\delta,m} + m\delta,$$

and set $\mathcal{A}_{\delta,m}(u) = \{S_u^{\delta,m} < \infty\}$ and $p_{\delta,m}(u) = \mathbb{P}(\mathcal{A}_{\delta,m}(u))$. This is a run of $m+1$ consecutive observations spanning the physical duration $m\delta$. It imposes no condition between observations. The cases $h = 0$ and $m = 0$ are ordinary simultaneous ruin under continuous and discrete observation, respectively. All windows and meshes are fixed as $u \rightarrow \infty$, unless a separate limit of constants is specified.

Set

$$(1.5) \quad a = \sqrt{c^2 + 2\beta}, \quad \kappa = c + a, \quad s_u = \sqrt{u/a^3}.$$

Write τ for the original ancestor's split time. Its first split creates a cohort of k_1 siblings at one point, with covariance R_1 . Start a copy of this cohort at zero and continue with the prescribed offspring and covariance sequence from its second entry. Let

$$(1.6) \quad L(s) = \min_{v \text{ alive in this cohort at } s} X_v(s) - cs, \quad M = \sup_{s \geq 0} L(s), \quad H(x) = \mathbb{P}(M > x), \quad x \geq 0.$$

The entire cohort is independent of (τ, B_0) : chronological indexing after the first split starts at the fixed next sequence entry and all future clocks are measured relative to their births. Let W be a standard Brownian motion independent of the cohort and define

$$(1.7) \quad Z(s) = \begin{cases} W(-s) - a(-s), & s \leq 0, \\ L(s), & s \geq 0. \end{cases}$$

This field is continuous at zero. Its forward part includes every subsequent split and covariance change.

For $h \geq 0$ and $0 \leq \theta < \delta$ put

$$(1.8) \quad \Gamma_h(Z) = \sup_{s \in \mathbb{R}} \inf_{0 \leq r \leq h} Z(s+r),$$

$$(1.9) \quad V_{\delta,m,\theta}(Z) = \sup_{j \in \mathbb{Z}} \min_{\ell=0,\dots,m} Z((j+\ell)\delta - \theta).$$

The windows can lie on either side of zero or straddle zero. A positive window may give a negative functional value, despite $Z(0) = 0$. Define

$$(1.10) \quad C_{0,h} = \frac{\beta}{a\kappa} \mathbb{E} \left\{ e^{\kappa \Gamma_h(Z)} \right\},$$

$$(1.11) \quad C_{\delta,m} = \frac{\beta}{a\kappa\delta} \int_0^\delta \mathbb{E} \left\{ e^{\kappa V_{\delta,m,\theta}(Z)} \right\} d\theta.$$

The expectation averages the future genealogy; the integral also averages the observation phase. Dependence on $\mathcal{E}$ is suppressed in the notation.

Theorem 1.1. *Consider the branching model above, with $c, \beta > 0$ and the bounded offspring and covariance sequence satisfying (1.1). For every fixed $h \geq 0$, $\delta > 0$ and $m \in \mathbb{N}_0$, the constants in (1.10)–(1.11) are finite and strictly positive, and*

$$(1.12) \quad p_{0,h}(u) \sim C_{0,h} e^{-\kappa u}, \quad p_{\delta,m}(u) \sim C_{\delta,m} e^{-\kappa u}.$$

The exponential rate depends only on the boundary slope and the individual splitting rate. The offspring and covariance sequences affect the constants through the forward cohort L . The upper offspring bound ensures nonexplosion; the moment estimate in Lemma A.1 is independent of that bound.

2. DISCUSSION AND FURTHER RESULTS

The same conditioning argument gives a normal approximation for the first successful window. Its start and completion are both concentrated near u/a , on the scale $s_u = \sqrt{u/a^3}$.

Corollary 2.1. *Under the assumptions of Theorem 1.1, consider either fixed observation rule. Denote its successful-window event by $\mathcal{A}(u)$, its start and completion by S_u, T_u , and its constant by C . Then, with G standard normal,*

$$(2.1) \quad \left(\frac{\tau - u/a}{s_u}, \frac{S_u - u/a}{s_u}, \frac{T_u - u/a}{s_u} \right) \Big| \mathcal{A}(u) \implies (G, G, G).$$

In particular, for $U_u = S_u$ or T_u ,

$$(2.2) \quad \sup_{t \geq 0} \left| e^{\kappa u} \mathbb{P}(U_u \leq t) - C \Phi \left(\frac{t - u/a}{s_u} \right) \right| \longrightarrow 0,$$

where Φ is the standard normal distribution function.

Thus, for large u , the completion-time approximation is

$$\mathbb{P}(T_u \leq t) \approx C e^{-\kappa u} \Phi \left(\frac{t - u/a}{\sqrt{u/a^3}} \right), \quad t \geq 0.$$

Equation (2.2) gives a uniform absolute error of $o(e^{-\kappa u})$. It does not assert a relative error bound in the far tails of the normal distribution.

Proposition 2.2. *The function $h \mapsto C_{0,h}$ is continuous and nonincreasing on $[0, \infty)$, and*

$$(2.3) \quad \lim_{h \downarrow 0} C_{0,h} = C_{0,0}, \quad \lim_{h \rightarrow \infty} C_{0,h} = 0.$$

For each fixed $\delta > 0$, $C_{\delta,m}$ is nonincreasing in m and converges to zero as $m \rightarrow \infty$. If $\delta_n \downarrow 0$ and $m_n \in \mathbb{N}_0$ satisfy $m_n \delta_n \rightarrow h \geq 0$, then

$$(2.4) \quad C_{\delta_n, m_n} \rightarrow C_{0,h}.$$

For ordinary ruin one additionally has

$$(2.5) \quad 0 < C_{\delta,0} < C_{0,0}, \quad C_{0,0} > 1, \quad \lim_{\delta \rightarrow \infty} C_{\delta,0} = 0,$$

and $C_{q\delta,0} \leq C_{\delta,0}$ for each positive integer q .

The limit (2.4) preserves physical duration by letting the number of grid steps grow when $h > 0$. Keeping m fixed instead yields $h = 0$. Continuous and sampled Parisian events of the same positive duration need not be nested: sampling can miss recoveries between observations. No general ordering of $C_{\delta,m}$ and $C_{0,m\delta}$ is asserted. Nor does convergence of constants alone assert a joint high-threshold theorem with a mesh depending on u . The bound $C_{0,0} > 1$ is specific to ordinary ruin and cannot persist for all positive window lengths in view of (2.3).

Proposition 2.3. *In the setting of Theorem 1.1, let $H(x) = \mathbb{P}(M > x)$ as in (1.6). Then*

$$(2.6) \quad C_{0,0} = 1 + \frac{\beta}{a} \int_0^\infty (e^{\kappa x} - e^{-(a-c)x}) H(x) dx$$

$$(2.7) \quad = \frac{(a-c)\mathbb{E}\{e^{\kappa M}\} + \kappa\mathbb{E}\{e^{-(a-c)M}\}}{2a}.$$

For every $u \geq 0$ one has the exact identity

$$(2.8) \quad C_{0,0} - e^{\kappa u} p_{0,0}(u) = \frac{\beta}{a} \int_u^\infty e^{\kappa x} (1 - e^{-2a(x-u)}) H(x) dx.$$

In particular, put $\gamma_ = c + \sqrt{c^2 + 4\beta}$ and fix $\kappa < r < \gamma_*$. With the constant D_r from Lemma A.1,*

$$(2.9) \quad 0 \leq C_{0,0} e^{-\kappa u} - p_{0,0}(u) \leq \frac{\beta D_r}{a} \left(\frac{1}{r - \kappa} - \frac{1}{r + a - c} \right) e^{-ru}.$$

The bound is uniform over all admissible environments.

Thus ordinary ruin has an exponentially smaller absolute remainder. The identity still involves the cohort tail H ; it is not an elementary formula for the genealogy-dependent constant. The offspring numbers and correlation matrices can also be specified by an exogenous random environment.

Remark 2.4. Let the sequence $\mathcal{E} = ((k_i, A_i))_{i \geq 1}$ be jointly random, independent of the lifetimes and Gaussian drivers, and satisfy (1.1) almost surely with a deterministic bound K . No independence between sequence entries, or between k_i and A_i , is required. The conclusions of Theorem 1.1 and Corollary 2.1 hold conditionally on $\mathcal{E}$. Without conditioning, they hold with constants $\mathbb{E}\{C_{0,h}(\mathcal{E})\}$ and $\mathbb{E}\{C_{\delta,m}(\mathcal{E})\}$.

Indeed, the conditional assertions follow from the deterministic results. The common choice of endpoint envelope D in Lemma A.2 bounds $e^{\kappa u} p(u \mid \mathcal{E})$ by $\int_{\mathbb{R}} D(x) dx$, independently of the environment. Dominated convergence therefore averages the probability equivalent and (3.9). The uniform remote-crossing bound in Lemma A.3 then transfers the averaged split-time limit to both window times.

We next consider continuous branch ruin. Throughout the branch-ruin results below, assume $k_i \equiv k \geq 2$ and $A_i = I_k$ for every i , and write

$$r = \beta(k-1), \quad c > \sqrt{2r}, \quad d = \sqrt{c^2 - 2r}, \quad \lambda = c + d.$$

For a particle v alive at time T , let $X_{v,T}^{\text{anc}}(s)$ denote its ancestral position at time $s \leq T$. Set

$$p_{\text{br},h}(u) = \mathbb{P} \left\{ \exists t \geq 0, v \in N(t+h) : \inf_{0 \leq s \leq h} (X_{v,t+h}^{\text{anc}}(t+s) - c(t+s)) > u \right\}, \quad h \geq 0.$$

Splits during the window are allowed, with the path continuing through a descendant. At $h = 0$ this is ordinary branch ruin. For $h > 0$, a moving infimum of the upper population envelope would permit switching lineages and would define a different event.

The first-hitting line gives an exact reduction for every fixed k . Let K_u be the number of particles absorbed when each ancestral line is stopped at its first hit of the drifted level u , and write $\omega_s^{(k)}(u) = \mathbb{E} \{s^{K_u}\}$, with $0^0 = 1$ and $K_0 = 1$.

Proposition 2.5. *Put $q_0(u) = p_{\text{br},0}(u)$ and $\pi_h = p_{\text{br},h}(0)$. For every $h \geq 0$, $\pi_h \in (0, 1]$. The function q_0 is continuous and strictly decreasing from 1 to 0 on $[0, \infty)$. There is a unique $a_h \geq 0$ such that $q_0(a_h) = \pi_h$, and*

$$(2.11) \quad p_{\text{br},h}(u) = 1 - \omega_{1-\pi_h}^{(k)}(u),$$

$$(2.12) \quad p_{\text{br},h}(u) = p_{\text{br},0}(u + a_h), \quad u \geq 0.$$

Moreover, for every fixed $h \geq 0$,

$$p_{\text{br},h}(u) \sim B_{k,h} e^{-\lambda u}, \quad B_{k,h} = B_{k,0} e^{-\lambda a_h}, \quad 0 < B_{k,0} < 1.$$

As $h \downarrow 0$, $a_h \downarrow 0$. In particular, for every deterministic $h(u) \geq 0$ tending to zero,

$$p_{\text{br},h(u)}(u) \sim p_{\text{br},0}(u) \sim B_{k,0} e^{-\lambda u}.$$

The constants also admit a common occupation representation. Construct a genealogical cluster $\mathcal{K}$ whose backward spine is $\xi_s = W_s - ds$, placed at relative time $-s$, $s \geq 0$. At independent rate- $k\beta$ Poisson points on this spine, attach $k-1$ independent ordinary BBM families with drift $-c$, evolving forwards. Attach one further ordinary family at time zero and include the backward spine itself. All grafted families are independent of the spine and the Poisson points. Let $\mathcal{O}(x)$ be the total particle-time the cluster spends above $-x$, and let $\Gamma_h^{\text{br}}(\mathcal{K})$ be the supremum, over compatible ancestral segments of length h , of their minimum position. Segments may cross a split or relative time zero.

Proposition 2.6. *For every $x > 0$, $0 < \mathcal{O}(x) < \infty$ almost surely. If E_λ is an independent exponential variable of rate λ , then*

$$(2.13) \quad p_{\text{br},h}(u) \sim B_{k,h} e^{-\lambda u}, \quad B_{k,h} = \frac{1}{\lambda d} \mathbb{E} \left\{ \frac{\mathbf{1}_{\{\Gamma_h^{\text{br}}(\mathcal{K}) > -E_\lambda\}}}{\mathcal{O}(E_\lambda)} \right\}, \quad h \geq 0.$$

The cluster and its occupation denominator are the same for all h ; only the numerator depends on the delay. At $h = 0$ the numerator is one. The constants are positive and nonincreasing, and satisfy

$$(2.14) \quad B_{k,h} \uparrow B_{k,0} \quad (h \downarrow 0), \quad B_{k,h} \leq \frac{1}{\lambda dh} \quad (h > 0), \quad B_{k,h} \rightarrow 0 \quad (h \rightarrow \infty).$$

The proofs are given in Section 4. For $k = 2$, the ordinary equivalent agrees, after reflection, with [7, Proposition 4]. That result also covers the critical speed, whereas the occupation representation here assumes $d > 0$. For fixed homogeneous k , the ordinary equivalent can also be obtained from the maximum-recursion theorem of [11, Theorem 3.4]. The proof below derives it directly from the branching equation and the additive martingale.

Corollary 2.7. *For $k = 2$, define*

$$(2.15) \quad \begin{aligned} \mathcal{S}(z) &= \sum_{n=1}^{\infty} a_n z^n, \quad a_1 = 1, \\ a_n &= \frac{\beta \sum_{j=1}^{n-1} a_j a_{n-j}}{(n-1)(n\lambda^2/2 - \beta)}, \quad n \geq 2. \end{aligned}$$

Then

$$(2.16) \quad p_{\text{br},h}(u) = \mathcal{S}(B_{2,h}e^{-\lambda u}) = \sum_{n=1}^{\infty} a_n B_{2,h}^n e^{-n\lambda u}, \quad u \geq 0, \quad h \geq 0.$$

Here $0 < B_{2,h} \leq B_{2,0} < R_{\mathcal{S}}$, where $R_{\mathcal{S}}$ is the convergence radius, and $\mathcal{S}(B_{2,0}) = 1$.

Proof. This is the series of [7, Proposition 9], applied with $s = 1 - \pi_h$ in (2.11). Its leading coefficient is one, so Proposition 2.5 identifies its argument constant with $B_{2,h}$. The same cited proposition gives $B_{2,0} < R_{\mathcal{S}}$ and convergence for every $u \geq 0$. $\square$

The fresh-family reduction above uses homogeneous offspring and independent sibling motions. The following example shows that bounded prescribed offspring alone need not give even a single branch-ruin decay rate.

Remark 2.8. There is a deterministic chronological sequence $k_i \in \{2, 3\}$, with $A_i = I_{k_i}$, $\beta = 1$ and $c = 3$, such that, for every fixed $h \geq 0$, the quantity

$$-\frac{1}{u} \log p_{\text{br},h}(u)$$

has $3 + \sqrt{7}$ and $3 + \sqrt{5}$ as distinct subsequential limits. Thus arbitrary bounded offspring sequences need not admit even a single logarithmic decay rate for classical or continuous Parisian branch ruin. Additional assumptions are needed to extend (2.13) to varying offspring; this does not affect Theorem 1.1 for simultaneous ruin. The construction, using alternating binary and ternary blocks, is proved at the end of Appendix A.

The first-hitting reduction concerns the existence of one successful ancestral line. In simultaneous ruin, every particle alive must satisfy the constraint on one common interval. The trees issued from the first-hitting line start at different absolute times, and their successful windows need not coincide. Thus K_u and π_h do not retain the timing and population information required for simultaneous ruin; the conditioning in Theorem 1.1 retains it through the common-window functional.

3. PROOFS FOR SIMULTANEOUS RUIN

We first record the endpoint identity and then use a direct bridge coupling for the full conditional window functional. The required uniform moment, kernel and phase estimates are proved in Appendix A.

Let br_t be a centered Brownian bridge on $[0, t]$, independent of the fresh cohort. For $t, u > 0$ and $x \in \mathbb{R}$ set

$$(3.1) \quad Z_{u,t,x}(s) = \begin{cases} \text{br}_t(-s) + (u-x)s/t, & -t \leq s \leq 0, \\ L(s), & s \geq 0. \end{cases}$$

For the two observation rules define

$$(3.2) \quad Q_{u,t}^{0,h}(x) = \mathbb{P} \left\{ \sup_{s \geq -t} \inf_{0 \leq r \leq h} Z_{u,t,x}(s+r) > x \right\},$$

$$(3.3) \quad Q_{u,t}^{\delta,m}(x) = \mathbb{P} \left\{ \sup_{j \in \mathbb{N}_0} \min_{\ell=0,\dots,m} Z_{u,t,x}((j+\ell)\delta - t) > x \right\}.$$

The lower restriction on starts excludes observation before absolute time zero. Write $Q_{u,t}^{\text{ord}} = Q_{u,t}^{0,0}$.

For $d > 0$, write

$$(3.4) \quad S_d = \sup_{s \geq 0} (W(s) - ds),$$

where W is the Brownian motion in (1.7), independent of the cohort.

Lemma 3.1. *For either rule, write $Q_{u,t}(x)$ for its conditional probability and $\mathcal{A}(u)$ for its event. For every bounded measurable f ,*

$$(3.5) \quad \mathbb{E} \{f(\tau); \mathcal{A}(u)\} = \int_{\mathbb{R}} \int_0^\infty f(t) \beta e^{-\beta t} \phi_t(u+ct-x) Q_{u,t}(x) dt dx, \quad \phi_t(z) = \frac{e^{-z^2/(2t)}}{\sqrt{2\pi t}}.$$

Moreover

$$(3.6) \quad Q_{u,t}(x) \leq Q_{u,t}^{\text{ord}}(x) = \mathbb{P}\{\max(S_{u/t}, M) > x\},$$

where the Brownian motion defining $S_{u/t}$ is independent of M .

Proof. Condition on $\tau = t$ and $B_0(t) - ct = u - x$. Gaussian regression gives

$$B_0(t-r) - c(t-r) - (B_0(t) - ct) \stackrel{d}{=} \text{br}_t(r) - (u-x)r/t, \quad 0 \leq r \leq t.$$

The independent forward cohort contributes $L(s)$. The complete conditional envelope, measured from its value at the split, is therefore (3.1). Applying the window definitions and integrating the endpoint proves (3.5) by Tonelli for nonnegative f , and then by subtraction for bounded f .

Every successful window contains an ordinary crossing, proving the inequality in (3.6). The bridge identity

$$(3.7) \quad \{\text{br}_t(r) : 0 \leq r < t\} \stackrel{d}{=} \left\{ (1-r/t)W\left(\frac{r}{1-r/t}\right) : 0 \leq r < t \right\}$$

follows by comparing Gaussian covariances. Substituting $v = r/(1-r/t)$ transforms a backward crossing into

$$\text{br}_t(r) - (u-x)r/t > x \iff W(v) - (u/t)v > x.$$

The cancellation is valid for every real x ; the endpoint $r = t$ has value $x - u < x$ and cannot cross. Combining this event with the forward maximum M gives the equality for $x \geq 0$. For $x < 0$ both ordinary events are certain because relative time zero is included. This last observation applies only to the ordinary envelope, not to a positive Parisian window. $\square$

Proof of Theorem 1.1. Using the endpoint identity of Lemma 3.1, we first prove a conditional approximation for the full window event. Write

$$\mathcal{F}_\theta(Z) = V_{\delta,m,\theta}(Z), \quad \mathcal{H}(\theta) = \mathbb{E} \left\{ e^{\kappa \mathcal{F}_\theta(Z)} \right\}$$

for the grid; for continuous observation use $\mathcal{F}(Z) = \Gamma_h(Z)$ and $\mathcal{H} = \mathbb{E} \left\{ e^{\kappa \mathcal{F}(Z)} \right\}$, and suppress the phase. Initially allow any $\theta \in [0, \delta]$ in the conditional grid field, retaining only lattice starts $j\delta - \theta \geq -t$, $j \in \mathbb{Z}$. Denote this conditional window probability by $\widehat{Q}_{u,t,\theta}(x)$. The actual conditional probability is $Q_{u,t}(x) = \widehat{Q}_{u,t,t \bmod \delta}(x)$.

On one probability space, independently of the forward cohort, realize the Brownian bridge by

$$\text{br}_t(r) = W(r) - \frac{r}{t}W(t), \quad 0 \leq r \leq t.$$

Thus the backward part of (3.1) has the law of

$$Z_{u,t,x}(-r) = W(r) - d_{u,t,x}r, \quad d_{u,t,x} = \frac{u - x + W(t)}{t}.$$

For every fixed $x \in \mathbb{R}$ and $0 < \Lambda < \infty$, the Brownian strong law gives, almost surely,

$$\sup_{|t-u/a| \leq \Lambda s_u} |d_{u,t,x} - a| \longrightarrow 0.$$

Eventually all these drifts are at least $a/2$. The backward fields are then bounded above by $W(r) - (a/2)r$, which tends to $-\infty$; the forward field L also tends to $-\infty$ by (A.2). A window starting at zero supplies the common lower bound $\inf_{0 \leq r \leq h} L(r)$ in the continuous case. For the grid, the first nonnegative lattice start lies in $[0, \delta]$, so the common lower bound $\inf_{0 \leq r \leq (m+1)\delta} L(r)$ works for every phase. Any start attaining at least this lower bound must therefore lie in one random compact interval independent of u , t and the phase, once u is sufficiently large (for the fixed x and Λ). The restriction on starts to $[-t, \infty)$ is then irrelevant to the maximum. Uniform convergence on that compact interval proves almost sure convergence of the full conditional window functionals to $\mathcal{F}_\theta(Z)$, uniformly in the central time range and in θ . The same compactness and path continuity show that $\theta \mapsto \mathcal{F}_\theta(Z)$ is continuous. It is δ -periodic by reindexing the lattice. The bound (A.3) consequently makes $\mathcal{H}(\theta)$ continuous and δ -periodic as well.

We justify integrating this pathwise convergence over the conditioned endpoint. Put

$$v = u + ct, \quad g_{u,t} = v/\sqrt{t}, \quad \rho_{u,t} = v/t.$$

Uniformly for $|t - u/a| \leq \Lambda s_u$,

$$\frac{\phi_t(v - x)}{\Psi(g_{u,t})} = \frac{\phi(g_{u,t})}{\sqrt{t} \Psi(g_{u,t})} e^{\rho_{u,t}x - x^2/(2t)}, \quad \frac{\phi(g_{u,t})}{\sqrt{t} \Psi(g_{u,t})} \longrightarrow \kappa,$$

where ϕ and Ψ are the standard normal density and survival function. Choose $r > \kappa$ allowed by Lemma A.1, and then choose $\varepsilon > 0$ with $\kappa - \varepsilon > 0$ and $\kappa + \varepsilon < r$. For all sufficiently large u in this time range,

$$\kappa - \varepsilon \leq \rho_{u,t} \leq \kappa + \varepsilon, \quad \frac{u}{t} - c \geq \frac{a - c}{2}.$$

Every phased grid event is contained in the ordinary conditional event, just as for continuous observation. Hence (3.6) bounds its conditional probability by $H(x) + e^{-2ux/t}$ for $x \geq 0$, and by one for $x < 0$. Lemma A.1 therefore gives the common integrable bound

$$\frac{\phi_t(v - x)}{\Psi(g_{u,t})} \widehat{Q}_{u,t,\theta}(x) \leq C_\Lambda \begin{cases} e^{(\kappa - \varepsilon)x}, & x < 0, \\ D_r e^{-(r - \kappa - \varepsilon)x} + e^{-(a - c)x/2}, & x \geq 0, \end{cases}$$

and the identical bound holds with the phase suppressed.

Along any sequence of central times and phases $\theta_u \rightarrow \theta$, the coupled window functionals converge almost surely to $\mathcal{F}_\theta(Z)$. Their exceedance probabilities thus converge for every x outside the at most countable set of atoms of this limiting functional. Dominated convergence and the layer-cake identity give

$$\int_{\mathbb{R}} \kappa e^{\kappa x} \mathbb{P}\{\mathcal{F}_\theta(Z) > x\} dx = \mathbb{E} \left\{ e^{\kappa \mathcal{F}_\theta(Z)} \right\}.$$

The integral includes negative endpoints, as required for positive windows. Compactness of the phase interval and continuity of $\mathcal{H}$ now give the uniform approximation

$$(3.8) \quad \sup_{|t-u/a| \leq \Lambda s_u} \left| \frac{\mathbb{P}(\mathcal{A}(u) \mid \tau = t)}{\Psi((u+ct)/\sqrt{t})} - \mathcal{H}(t \bmod \delta) \right| \rightarrow 0.$$

For continuous observation the last term is $\mathcal{H}$. In the grid case, a sequence violating uniformity would have a subsequence with convergent phases in $[0, \delta]$. The preceding sequential convergence and continuity of $\mathcal{H}$ give a contradiction; periodicity identifies the endpoint phase δ with zero. The continuous case uses the same argument without a phase.

Let f be bounded and continuous, and set $y_u(t) = (t - u/a)/s_u$. On $|y_u(t)| \leq \Lambda$, uniformly,

$$\Psi((u+ct)/\sqrt{t}) \sim \frac{1}{\kappa} \phi_t(u+ct), \quad e^{\kappa u} \beta e^{-\beta t} \phi_t(u+ct) dt = \frac{\beta}{a} \nu_u(dt).$$

Apply (3.8) and the normal and phase limits in Lemmas A.2 and A.4. They yield

$$e^{\kappa u} \mathbb{E} \{ f(y_u(\tau)); \mathcal{A}(u), |y_u(\tau)| \leq \Lambda \} \rightarrow \frac{\beta}{a\kappa} \overline{\mathcal{H}} \mathbb{E} \{ f(G); |G| \leq \Lambda \},$$

where $\overline{\mathcal{H}} = \mathcal{H}$ for continuous observation and $\overline{\mathcal{H}} = \delta^{-1} \int_0^\delta \mathcal{H}(\theta) d\theta$ for the grid. Since $\mathcal{A}(u) \subset \mathcal{A}_{\text{ord}}(u)$, the split-time localization (A.7) removes the restriction $|y_u(\tau)| \leq \Lambda$ as $\Lambda \rightarrow \infty$. By the definitions of the constants, $\beta \overline{\mathcal{H}}/(a\kappa) = C$. Consequently

$$(3.9) \quad e^{\kappa u} \mathbb{E} \left\{ f\left(\frac{\tau - u/a}{s_u}\right); \mathcal{A}(u) \right\} \rightarrow C \mathbb{E} \{ f(G) \}.$$

Taking $f = 1$ proves (1.12); positivity and finiteness follow from (A.3) and the finite pathwise window values. $\square$

Proof of Corollary 2.1. Dividing (3.9) by the probability equivalent proves the conditional normal limit for τ . Moreover,

$$\{|S_u - \tau| > R\} \cap \mathcal{A}(u) \subset E_R(u).$$

For continuous observation, the infimum defining S_u need not itself start a strict successful window. Successful starts approach it, and the strict inequality $|S_u - \tau| > R$ still supplies an ordinary crossing farther than R from τ . For the grid the first successful start is attained. Lemma A.3 therefore gives

$$\lim_{R \rightarrow \infty} \limsup_{u \rightarrow \infty} \mathbb{P}\{|S_u - \tau| > R \mid \mathcal{A}(u)\} = 0.$$

Since $s_u \rightarrow \infty$ and the completion adds the fixed duration h or $m\delta$, both $(S_u - \tau)/s_u$ and $(T_u - \tau)/s_u$ converge to zero in conditional probability. Slutsky's theorem proves (2.1). The normal limit has a continuous distribution function, so convergence of the conditional distribution functions is uniform. Combining this with the probability equivalent proves (2.2). $\square$

Proof of Proposition 2.2. Consider first a deterministic continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(s) \rightarrow -\infty$ as $|s| \rightarrow \infty$. For $0 \leq h \leq H_0 < \infty$,

$$\Gamma_h(f) \geq b := \inf_{0 \leq s \leq H_0} f(s).$$

Any maximizing start must therefore belong to the compact set $\{f \geq b\}$, since the window includes its start. On a compact interval containing these starts and all their windows, uniform continuity gives

$$|\Gamma_h(f) - \Gamma_{h'}(f)| \leq \omega_f(|h - h'|), \quad 0 \leq h, h' \leq H_0,$$

where ω_f is the modulus of continuity on that interval. This proves continuity in h . Monotonicity is immediate. For any $b \in \mathbb{R}$, the bounded set $\{f \geq b\}$ cannot contain intervals of arbitrarily large length. Hence $\Gamma_h(f) \rightarrow -\infty$ as $h \rightarrow \infty$. Similarly, at fixed grid and phase the sampled set above any fixed level is finite, so $V_{\delta, m, \theta}(f) \rightarrow -\infty$ as $m \rightarrow \infty$; monotonicity in m is immediate.

For the dense-grid assertion, suppose $\delta_n \rightarrow 0$ and $m_n \delta_n \rightarrow h$. A grid start within δ_n of zero supplies a common finite lower bound for $V_{\delta_n, m_n, \theta}(f)$, uniformly in $0 \leq \theta < \delta_n$, for all sufficiently large n . At a maximizing start the first sampled value must exceed or equal this bound. Thus all maximizing starts, and their windows, lie in a single compact interval independent of n and θ . Include also the maximizing starts for $\Gamma_h(f)$, and enlarge the interval slightly. Mapping a continuous start to a nearest grid start, and comparing the continuous window with its sampled points, gives

$$(3.10) \quad \sup_{0 \leq \theta < \delta_n} |V_{\delta_n, m_n, \theta}(f) - \Gamma_h(f)| \leq \omega_f(2\delta_n + |m_n \delta_n - h|) \rightarrow 0.$$

Indeed, the starts differ by at most δ_n , and the Hausdorff distance between the resulting window sets is at most $2\delta_n + |m_n \delta_n - h|$. Comparing minima on these sets and then suprema over starts proves both inequalities in (3.10).

Apply these pathwise facts to $f = Z$. All exponentials are bounded by $e^{\kappa \sup Z}$, which is integrable by (A.3). Dominated convergence proves continuity and (2.3), the fixed-grid limit as $m \rightarrow \infty$, and (2.4), the latter after writing $\theta = \delta_n v$, $0 \leq v < 1$.

For the ordinary comparisons, take U uniform on $(0, 1)$ independently of Z . Always $V_{\delta, 0, \delta U}(Z) \leq \sup Z$. Independence and the exponential law of S_a give

$$\mathbb{P}(S_a > M) = \mathbb{E} \{e^{-2aM}\} > 0,$$

since M is finite almost surely. On this event the global maximum is the almost surely unique maximum of a Brownian motion with negative drift, located at a strictly negative time. An independent uniform lattice phase misses this point almost surely. The sampled maximum is attained, by (A.2), so it is strictly smaller on this event. This proves $C_{\delta, 0} < C_{0, 0}$.

As $\delta \rightarrow \infty$, every point of $\delta\mathbb{Z} - \delta U$ has absolute value at least $\delta \min(U, 1 - U) \rightarrow \infty$. Equation (A.2) and (A.3) therefore give $C_{\delta, 0} \rightarrow 0$. For a positive integer q , let Θ be uniform on $[0, q\delta)$. The lattice $q\delta\mathbb{Z} - \Theta$ is contained in $\delta\mathbb{Z} - \Theta$, and $\Theta \bmod \delta$ is uniform on $[0, \delta)$. This proves $C_{q\delta, 0} \leq C_{\delta, 0}$.

The inequality $C_{0, 0} > 1$ follows from (2.7) and the support argument in the proof of Proposition 2.3 below. $\square$

Proof of Proposition 2.3. Independence and the exponential law of S_a give, for $m \geq 0$,

$$\mathbb{E} \left\{ e^{\kappa \max(S_a, m)} \right\} = e^{\kappa m} + \frac{\kappa}{2a - \kappa} e^{-(2a - \kappa)m}.$$

Since $2a - \kappa = a - c$ and $\kappa(a - c) = 2\beta$, averaging over M in (1.10) proves (2.7). Tail integration proves (2.6).

For $x \geq 0$, the exact conditional ordinary-crossing probability is

$$Q_{u,t}^{\text{ord}}(x) = H(x) + (1 - H(x))e^{-2ux/t}.$$

For $x < 0$ it equals one. Substitution in (3.5), followed by (A.4) and (A.8), yields, for $u > 0$,

$$\begin{aligned} e^{\kappa u} p_{0,0}(u) &= \frac{\beta}{a\kappa} + \frac{\beta}{a} \int_0^u e^{\kappa x} H(x) dx + \frac{\beta}{a} e^{2au} \int_u^\infty e^{-(a-c)x} H(x) dx \\ &\quad + \frac{\beta}{a} \int_0^\infty e^{-(a-c)x} (1 - H(x)) dx. \end{aligned}$$

Use $\beta/(a\kappa) + \beta/[a(a-c)] = 1$ and subtract this expression from (2.6). This proves (2.8). The case $u = 0$ follows from $p_{0,0}(0) = 1$, since the initial Brownian path crosses zero before its strictly positive first split time almost surely. Substituting $H(x) \leq D_r e^{-rx}$ and integrating gives (2.9).

For completeness, the function

$$m \mapsto \frac{(a-c)e^{\kappa m} + \kappa e^{-(a-c)m}}{2a}$$

equals one at zero and is strictly increasing for $m > 0$. We show that $\mathbb{P}(M > 0) > 0$, including singular sibling matrices. Require the k_1 initial siblings to split at ordered distinct times $0 < s_1 < \dots < s_{k_1} < 1$, and none of their children to split before time two. Conditional on these lifetimes, $(W_i^1(s_i))_{i=1}^{k_1}$ has positive-definite covariance: in a nonzero linear combination, the largest nonzero index j contributes an independent positive-variance increment on $(s_{j-1}, s_j]$, with $s_0 = 0$. All initial siblings therefore split above $2c + 2$ with positive probability. Independently, each fresh newborn vector has positive probability of having all its increments at time two in $(-1, 1)$, even when its covariance is singular. These finite clock restrictions also have positive probability, and together imply $L(2) > 0$. Equation (2.7) now proves $C_{0,0} > 1$. $\square$

4. PROOFS FOR CONTINUOUS BRANCH RUIN

Throughout this section, k is fixed, siblings move independently, $r = \beta(k-1)$, $d = \sqrt{c^2 - 2r} > 0$ and $\lambda = c + d$. Spine and cluster coordinates include the drift $-c$; $X_v(t)$ retains its original meaning as an undrifted Brownian position.

Proof of Proposition 2.5. We first establish the ordinary equivalent independently of the cluster representation. The additive process $\sum_{v \in N(t)} e^{\lambda(X_v(t) - ct)}$ is a nonnegative mean-one martingale, since $\lambda^2/2 - c\lambda + r = 0$. Its maximal inequality gives $q_0(u) \leq e^{-\lambda u}$. A single Brownian path can hit u before its first split, so $q_0(u) > 0$ and $q_0(u) \rightarrow 1$ as $u \downarrow 0$. Also $q_0(0) = 1$, by the immediate positive excursions of Brownian motion before that split.

For $0 < \ell < u < b < \infty$, let $D_t = u + B(t) + ct$ and let T_I be its first exit time from $I = (\ell, b)$. Conditioning on this exit or the first split gives the stopped-resolvent identity

$$\begin{aligned} q_0(u) &= \mathbb{E} \left\{ e^{-\beta T_I} q_0(D_{T_I}) \right\} \\ &\quad + \beta \mathbb{E} \left\{ \int_0^{T_I} e^{-\beta t} [1 - (1 - q_0(D_t))^k] dt \right\}. \end{aligned}$$

The exit term is smooth, and the killed one-dimensional Brownian resolvent with drift c maps bounded measurable sources to continuous functions. Thus q_0 is continuous on I ; the same

representation, now with continuous source, gives $q_0 \in C^2(I)$. Since I is arbitrary, the local branching equation is

$$\frac{1}{2}q_0'' + cq_0' + rq_0 = g(q_0), \quad g(z) = \beta[(1-z)^k - 1 + kz], \quad u > 0.$$

For $0 < z \leq 1$, $0 < g(z) \leq Cz^2$, where $C = \beta k(k-1)/2$. Set $v(u) = e^{\lambda u} q_0(u)$, so that $0 < v \leq 1$ and

$$\frac{1}{2}v'' - dv' = e^{\lambda u} g(q_0(u)).$$

The right side is bounded by $Ce^{-\lambda u}$. Solving the linear equation for v' and using boundedness of v to eliminate its exponentially growing solution yields

$$(4.1) \quad v'(u) = -2 \int_0^\infty e^{-2dt} e^{\lambda(u+t)} g(q_0(u+t)) dt < 0.$$

Thus v decreases to a limit $B_{k,0} < 1$, and q_0 is strictly decreasing. Using this monotonicity in the same integral gives

$$-v'(u) \leq \frac{2C}{2d + \lambda} e^{-\lambda u} v(u)^2.$$

Hence $(1/v)'$ is integrable on $(0, \infty)$, proving $B_{k,0} > 0$ and $q_0(u) \sim B_{k,0} e^{-\lambda u}$.

For completeness, the first-hitting line is almost surely finite. The first-moment calculation for a marked ancestral path gives

$$\mathbb{E} \{K_u\} = \int_0^\infty e^{rt} \frac{u}{\sqrt{2\pi t^3}} \exp\left\{-\frac{(u+ct)^2}{2t}\right\} dt = e^{-\lambda u}, \quad u > 0.$$

Indeed, at each split summing the k possible continuing children multiplies the path weight by k ; the resulting mean factor up to time t is e^{rt} . Stopping the marked path at its first boundary hit gives the Brownian first-hitting density displayed above. The identity follows first with bounded times and generation numbers and then by monotone convergence.

At the K_u boundary hits the future trees are independent fresh copies. Every successful window above u occurs after the first hit of u on its ancestral line. This proves (2.11). For each finite h , $\pi_h > 0$: there is positive probability of no split for a sufficiently long interval and of a Brownian segment staying strictly above zero for time h . Thus $\pi_h \in (0, 1]$, and the established properties of q_0 give a unique $a_h \geq 0$ with $q_0(a_h) = \pi_h$. Applying the same hitting-line decomposition to an ordinary crossing of $u + a_h$ gives

$$q_0(u + a_h) = 1 - \mathbb{E} \{(1 - q_0(a_h))^{K_u}\} = p_{\text{br},h}(u).$$

The ordinary equivalent now proves the asserted equivalent for each h . As $h \downarrow 0$, the events defining π_h increase to ordinary ruin at level zero: a strict crossing contains a strictly positive ancestral segment of some positive length. Consequently $\pi_h \uparrow 1$ and $a_h \downarrow 0$. Finally,

$$\frac{p_{\text{br},h(u)}(u)}{q_0(u)} = e^{-\lambda a_h(u)} \frac{v(u + a_h(u))}{v(u)} \longrightarrow 1,$$

which proves the assertion for vanishing delays. $\square$

Proof of Proposition 2.6. We first prove the representation for $h > 0$. For an ordinary family started at zero, let $U(z)$ be its expected total occupation above z . The first-moment identity and a Gaussian integral give

$$(4.2) \quad U(z) = \int_0^\infty e^{rt} \mathbb{P}\{B(t) - ct > z\} dt, \quad U(z) = \frac{e^{-\lambda z}}{\lambda d} \quad (z \geq 0).$$

The first integral is finite also for every negative z . To verify both assertions, integrate its spatial density:

$$\int_0^\infty e^{rt} \frac{1}{\sqrt{2\pi t}} e^{-(y+ct)^2/(2t)} dt = \frac{1}{d} e^{-cy-d|y|}, \quad y \in \mathbb{R}.$$

Let $A_{u,h}$ denote the branch event and let O_u be the total occupation of the original BBM above u . It is finite almost surely by (4.2); on $A_{u,h}$ it is positive. With the ratio defined as zero outside $\{0 < O_u < \infty\}$, Tonelli's theorem gives

$$(4.3) \quad p_{\text{br},h}(u) = \int_0^\infty \mathbb{E} \left\{ \sum_{v \in N(t)} \mathbf{1}_{\{X_v(t)-ct > u\}} \frac{\mathbf{1}_{A_{u,h}}}{O_u} \right\} dt.$$

Here and below these identities hold for any nonnegative measurable functional of the whole tree. To describe the marked-tree law in (4.3), mark one particle at time t and trace its ancestry. Summing the possible continuing children weights each ancestral split by k . Thus the ancestral split process has rate $k\beta$, with overall factor e^{rt} ; at each split the other $k-1$ children start independent ordinary families. The marked path itself remains Brownian with drift $-c$, and at time t its future is one further ordinary family. This identity follows directly by expanding over ancestral split times and child choices, initially for bounded tree functionals depending on finitely many generations and times. A monotone-class argument and monotone convergence extend it to the nonnegative functional in (4.3).

Condition that the marked position at time t equals $z = u + x$, $x > 0$, and translate that point to $(0, 0)$. The reversed ancestral path is

$$\xi_s^{t,z} = \text{br}_t(s) - \frac{z}{t}s, \quad 0 \leq s \leq t.$$

Let $\mathcal{K}_{t,z}$ be the tree formed from this finite backward spine, the rate- $k\beta$ grafts, and the fresh family at zero. All side families retain their entire future evolution. Define $\mathcal{O}_{t,z}(x)$ and $\Gamma_h^{\text{br}}(\mathcal{K}_{t,z})$ on this tree, whose earliest time is $-t$, and put

$$F_h(\mathcal{K}_{t,z}, x) = \frac{\mathbf{1}_{\{\Gamma_h^{\text{br}}(\mathcal{K}_{t,z}) > -x\}}}{\mathcal{O}_{t,z}(x)}.$$

For $h > 0$ this ratio lies in $[0, 1/h]$, since a successful ancestral segment contributes h units of occupation. With the normal density ϕ_t defined in (3.5), (4.3) becomes the exact identity

$$(4.4) \quad p_{\text{br},h}(u) = \int_0^\infty \int_0^\infty e^{rt} \phi_t(u+x+ct) \mathbb{E} \{F_h(\mathcal{K}_{t,u+x}, x)\} dt dx.$$

For $z > 0$, the probability measure

$$\mu_z(dt) = d e^{\lambda z} e^{rt} \phi_t(z+ct) dt = \frac{d}{\sqrt{2\pi t}} \exp\left\{-\frac{(z-dt)^2}{2t}\right\} dt$$

has mean $z/d + 1/d^2$ and variance $z/d^3 + 2/d^4$, as follows by differentiating its Laplace transform

$$\int e^{-qt} \mu_z(dt) = \frac{d}{\sqrt{d^2 + 2q}} \exp\{z(d - \sqrt{d^2 + 2q})\}, \quad q \geq 0.$$

In particular $t/z \rightarrow 1/d$ in μ_z -probability. Consequently

$$(4.5) \quad e^{\lambda u} p_{\text{br},h}(u) = \frac{1}{d} \int_0^\infty e^{-\lambda x} \int \mathbb{E} \{F_h(\mathcal{K}_{t,u+x}, x)\} \mu_{u+x}(dt) dx.$$

We next justify convergence of the full-tree functional in this formula. If $t \rightarrow \infty$ and $z/t \rightarrow d$, the finite reversed spine converges on every bounded interval to $W_s - ds$. The Poisson graft

times and the ordinary side families can be coupled identically on such an interval. Two truncation estimates allow passage from this local convergence to the full tree.

First consider grafts at backward ages $s \geq R$. Put $\eta = d/4$ and suppose $z/t \geq d/2$. The Brownian-bridge time change $\text{br}_t(s) \stackrel{d}{=} (1 - s/t)W_{s/(1-s/t)}$ implies

$$(4.6) \quad \mathbb{P}\{\exists s \in [R, t] : \xi_s^{t,z} > -\eta s\} \leq 2\Psi(d\sqrt{R}/4),$$

where Ψ is the standard normal survival function, as above. Indeed, after the time change the crossing is bounded by a crossing of $W_v - (d/4)v$ after time R , whose probability is the displayed normal-tail bound. Denote the complementary event by G_R . If $R > x/\eta$, the old spine contributes no occupation above $-x$ on G_R . Conditional on the spine, Campbell's formula for its independent Poisson grafts and (4.2) give

$$\mathbb{E}\{\mathcal{O}_{\text{old}}(x) \mid \xi^{t,z}\} \leq \frac{k\beta(k-1)}{\lambda^2 d\eta} e^{\lambda x - \lambda\eta R} \quad \text{on } G_R.$$

Here $\mathcal{O}_{\text{old}}$ includes the old spine and the entire future of every family grafted at age at least R . Therefore, for every $\varepsilon > 0$,

$$(4.7) \quad \mathbb{P}\{\mathcal{O}_{\text{old}}(x) > \varepsilon\} \leq 2\Psi(d\sqrt{R}/4) + \frac{k\beta(k-1)}{\varepsilon\lambda^2 d\eta} e^{\lambda x - \lambda\eta R}.$$

This is uniform in $t \geq R$ and $z/t \geq d/2$. The same bound holds for the infinite spine, with no larger first term. Moreover, its almost sure linear decrease and the conditional occupation estimate show that $\mathcal{O}(x) < \infty$ almost surely: only finitely many recent grafts remain, each with finite occupation. The fresh family at zero gives $\mathcal{O}(x) > 0$ when $x > 0$. On G_R , a successful segment omitted by the age truncation cannot use the old spine, which lies below $-x$. It must lie entirely in an old grafted family and hence contributes at least h to $\mathcal{O}_{\text{old}}(x)$. Thus the same estimate controls the truncation of the numerator.

Second, keep only grafts of age at most R , and truncate their future at relative time $S > h$. On the event that the retained spine positions are bounded in absolute value by L , an ordinary family started at any such position y satisfies

$$\mathbb{E}\left\{\int_T^\infty \sum_{v \in N(t)} \mathbf{1}_{\{y + X_v(t) - ct > -x\}} dt\right\} \leq \frac{2}{d^2} e^{c(x+L)} e^{-d^2 T/2}.$$

This follows from exponential Markov inequality with parameter c , using $r - c^2/2 = -d^2/2$. There are on average $1 + k\beta(k-1)R$ retained families, including the fresh family at zero. Their occupation after relative time S is therefore bounded in expectation, on that spine event, by this same factor times $(2/d^2)e^{c(x+L)}e^{-d^2 S/2}$. For the numerator use the bound with $S - h$: any successful segment ending after S contributes h occupation after $S - h$. The retained spines are tight on $[0, R]$ under $t \rightarrow \infty$, $z/t \rightarrow d$, so L can first be chosen large and then S large. These estimates prove uniform forward truncation for both functionals.

On a fixed truncation $[-R, S]$ the genealogy has finitely many edges almost surely. Under the coupling above, its positions converge uniformly edge by edge. The supremum of ancestral-window minima changes by at most the largest positional change. Occupation also converges, because Brownian paths and bridges spend zero Lebesgue time at a specified level. For Lebesgue-almost every $x > 0$ the limiting supremum of ancestral-window minima is not $-x$ almost surely: for each realization a random variable can equal $-x$ for at most one x , so this follows by Fubini. Take integer cutoffs R, S to use one common full-measure set of x . The occupation of the identically coupled fresh family on $[0, 1]$ is positive almost surely, since its

initial path starts strictly above $-x$. For $S \geq 1$ it supplies a common positive lower bound for all the occupation denominators. The two truncation estimates now give

$$\mathbb{E} \{F_h(\mathcal{K}_{t,z}, x)\} \longrightarrow \mathbb{E} \left\{ \frac{\mathbf{1}_{\{\Gamma_h^{\text{br}}(\mathcal{K}) > -x\}}}{\mathcal{O}(x)} \right\} \quad (t \rightarrow \infty, z/t \rightarrow d)$$

for almost every $x > 0$. One may first restrict this common lower bound away from zero and then remove that restriction; the bound $F_h \leq 1/h$ makes the latter step harmless. The concentration of μ_{u+x} and dominated convergence in (4.5), with dominating function $e^{-\lambda x}/h$, establish (2.13) for $h > 0$.

It remains to identify the zero-delay formula; no inverse-occupation domination is assumed. Since $E_\lambda > 0$ almost surely, continuity of the fresh family at zero makes $\mathbf{1}_{\{\Gamma_h^{\text{br}}(\mathcal{K}) > -E_\lambda\}} \uparrow 1$ as $h \downarrow 0$. By Proposition 2.5, independently of this representation, $B_{k,h} = B_{k,0}e^{-\lambda a_h} \uparrow B_{k,0}$. Monotone convergence applied to the already proved positive-delay formulas therefore gives

$$\frac{1}{\lambda d} \mathbb{E} \left\{ \frac{1}{\mathcal{O}(E_\lambda)} \right\} = B_{k,0} < \infty.$$

This proves the representation at $h = 0$. Positivity and monotonicity follow from Proposition 2.5; alternatively the fresh family has a positive chance of a successful window of any fixed length. Finally $F_h \leq 1/h$ gives $B_{k,h} \leq 1/(\lambda d h)$ and hence $B_{k,h} \rightarrow 0$ as $h \rightarrow \infty$. $\square$

APPENDIX A. TECHNICAL ESTIMATES

The estimates in this appendix do not assume any probability equivalent. They are uniform over the admissible offspring and covariance sequences.

The standard Brownian maximum identity gives, for $d > 0$,

$$(A.1) \quad \mathbb{P}(S_d > x) = e^{-2dx}, \quad x \geq 0.$$

Lemma A.1. *Let L be the lower envelope of a newborn cohort in the stated model, and let $M = \sup_{t \geq 0} L(t)$. Put*

$$\gamma_* = c + \sqrt{c^2 + 4\beta}.$$

For every $0 < r < \gamma_$ there is a finite constant D_r , depending only on c, β, r , such that*

$$\mathbb{P}\{M > u\} \leq D_r e^{-ru}, \quad u \geq 0.$$

In particular, since $\gamma_ > \kappa$, the moment $\mathbb{E}\{e^{\kappa M}\}$ is bounded uniformly over all admissible offspring and covariance sequences.*

Proof. Retain the first two initial siblings and, whenever a retained particle splits, retain its first two children. Give each retained initial sibling weight $1/2$, and divide a parent's weight equally between its two retained children. A retained particle v at depth $g \geq 1$ then has weight $w_v = 2^{-g}$. The retained labels form a fixed infinite binary tree, because every split has at least two children. In the usual construction with an independent lifetime assigned to every potential particle label, the lifetimes (τ_v) on this binary tree are therefore independent $\text{Exp}(\beta)$ variables. This remains true when the assignments at different splits use a common chronological index; no independence of the full descendant families is asserted or needed.

Let $\mathcal{B}(t)$ be the retained particles alive at time t and set

$$\bar{X}(t) = \sum_{v \in \mathcal{B}(t)} w_v X_v(t).$$

There are finitely many particles on compact time intervals by nonexplosion of the population. The living weights sum to one, and a split does not change $\bar{X}$, because both retained children start at their parent's position. Consequently $\bar{X}$ is continuous and

$$L(t) \leq \bar{X}(t) - ct, \quad t \geq 0.$$

Condition on all lifetime variables, denoting this information by $\mathcal{G}$. Birth times, retained weights and all covariance matrices are then fixed. Contributions from different birth events are independent, and increments from any one birth event over disjoint absolute time intervals use disjoint Brownian increments. Thus, conditional on $\mathcal{G}$, $\bar{X}$ is a centered continuous Gaussian process with independent increments. Write

$$V(t) = \text{Var}(\bar{X}(t) \mid \mathcal{G}).$$

For a birth event e , let $J_e(t)$ be its retained children still alive at time t , and let R^e be its sibling covariance matrix. Except at the finitely many split times in any compact interval,

$$V'(t) = \sum_e \sum_{i,j \in J_e(t)} w_{e,i} w_{e,j} R_{ij}^e \leq \sum_e \left(\sum_{i \in J_e(t)} w_{e,i} \right)^2 \leq 2 \sum_{v \in \mathcal{B}(t)} w_v^2.$$

Here $|J_e(t)| \leq 2$ and $R_{ij}^e \leq 1$. Initially the sum of squared weights is $1/2$; replacing weight w by two weights $w/2$ decreases that sum by $w^2/2$. Hence

$$0 \leq V'(t) \leq 1, \quad V(t) \leq t,$$

and integration gives

$$V(\infty) \leq J_\infty, \quad J_\infty := 2 \sum_{g \geq 1} 4^{-g} \sum_{|v|=g} \tau_v.$$

In particular $\mathbb{E}\{J_\infty\} = 2/\beta < \infty$. Independence of the retained lifetimes and monotone convergence yield, for $0 \leq \alpha < 2\beta$,

$$\mathbb{E}\{e^{\alpha J_\infty}\} = \prod_{g \geq 1} \left(1 - \frac{2\alpha 4^{-g}}{\beta}\right)^{-2g} < \infty.$$

To see convergence of the product, its logarithm is bounded by a constant times $\sum_{g \geq 1} 2^g 4^{-g} < \infty$, because the largest value of $2\alpha 4^{-g}/\beta$ is $\alpha/(2\beta) < 1$.

For $0 < r < \gamma_*$ put $\alpha_r = (r^2/2 - rc)_+ < 2\beta$. Conditional on $\mathcal{G}$,

$$\mathcal{M}_r(t) = \exp\{r\bar{X}(t) - r^2 V(t)/2\}$$

is a nonnegative mean-one martingale. Since $V(t) \leq t$ and $V(t) \leq J_\infty$,

$$r^2 V(t)/2 - rct \leq (r^2/2 - rc)V(t) \leq \alpha_r J_\infty.$$

The martingale maximal inequality, first on finite horizons and then by monotone convergence, yields

$$\mathbb{P}(M > u \mid \mathcal{G}) \leq \mathbb{P}\{\sup_{t \geq 0} \mathcal{M}_r(t) > e^{ru - \alpha_r J_\infty} \mid \mathcal{G}\} \leq e^{-ru + \alpha_r J_\infty}.$$

Thus $D_r = \mathbb{E}\{e^{\alpha_r J_\infty}\} < \infty$ is an admissible constant. Finally, choose $r \in (\kappa, \gamma_*)$. Since $M \geq 0$, tail integration gives the uniform estimate

$$\mathbb{E}\{e^{\kappa M}\} = 1 + \kappa \int_0^\infty e^{\kappa u} \mathbb{P}(M > u) du \leq 1 + \frac{\kappa D_r}{r - \kappa} < \infty.$$

□

Following one specified child at every split gives a standard Brownian lineage: conditional on the lifetimes its independent Gaussian segments have covariance $\min(s, t)$. Hence

$$(A.2) \quad L(s) \longrightarrow -\infty \quad (s \rightarrow \infty), \quad Z(s) \longrightarrow -\infty \quad (|s| \rightarrow \infty) \quad \text{almost surely.}$$

Since $\kappa < 2a$, (A.1) gives $\mathbb{E} \{e^{\kappa S_a}\} = 2a/(2a - \kappa) < \infty$. Together with Lemma A.1, this yields

$$(A.3) \quad \mathbb{E} \left\{ e^{\kappa \sup_{s \in \mathbb{R}} Z(s)} \right\} = \mathbb{E} \left\{ e^{\kappa \max(S_a, M)} \right\} < \infty.$$

This bound is uniform over $\mathcal{E}$. Every functional in (1.8)–(1.9) is finite, is attained, and is bounded above by $\sup Z$. For the continuous functional, the moving infimum is continuous, and it tends to $-\infty$ when its start tends to either infinity. For the sampled functional the same conclusion follows along the shifted lattice. Each functional is bounded below by its value at any single admissible start. This also proves positivity and finiteness of all the constants.

Lemma A.2. *For $z \in \mathbb{R}$,*

$$(A.4) \quad \int_0^\infty \frac{1}{\sqrt{2\pi t}} e^{-(z-at)^2/(2t)} dt = \frac{1}{a} e^{az-a|z|} \leq \frac{1}{a}.$$

For $z > 0$ the measure

$$(A.5) \quad \nu_z(dt) = \frac{a}{\sqrt{2\pi t}} e^{-(z-at)^2/(2t)} dt$$

is a probability measure. For each fixed x , the density under ν_{u-x} of $(t - u/a)/s_u$ converges in L^1 to the standard normal density.

There is an integrable $D : \mathbb{R} \rightarrow [0, \infty)$ such that, for every $u > 0$ and every conditional window probability in (3.2)–(3.3),

$$(A.6) \quad e^{\kappa u} \int_0^\infty \beta e^{-\beta t} \phi_t(u + ct - x) Q_{u,t}(x) dt \leq D(x).$$

The same function can be chosen for all admissible environments. Moreover, writing $\mathcal{A}_{\text{ord}}(u) = \mathcal{A}_{0,0}(u)$,

$$(A.7) \quad \lim_{\Lambda \rightarrow \infty} \limsup_{u \rightarrow \infty} e^{\kappa u} \mathbb{P}\{\mathcal{A}_{\text{ord}}(u), |\tau - u/a| > \Lambda s_u\} = 0.$$

Proof. The elementary Gaussian integral

$$\int_0^\infty \frac{1}{\sqrt{2\pi t}} e^{-A^2 t/2 - z^2/(2t)} dt = A^{-1} e^{-A|z|}, \quad A > 0,$$

gives (A.4). Put $t_u(y) = u/a + y s_u$. For $u > x$, the transformed density is

$$f_{u,x}(y) = \frac{a s_u}{\sqrt{2\pi t_u(y)}} e^{-(u-x-at_u(y))^2/(2t_u(y))} \mathbf{1}_{\{t_u(y) > 0\}},$$

with value zero when $t_u(y) \leq 0$. It converges pointwise to the standard normal density, and both densities integrate to one. Scheffé's lemma gives the asserted L^1 convergence. In particular, $t/u \rightarrow 1/a$ in ν_{u-x} -probability.

The square completion

$$(A.8) \quad \beta t + \frac{(u + ct - x)^2}{2t} = \kappa(u - x) + \frac{(u - x - at)^2}{2t}$$

is exact. For $x < 0$, use $Q_{u,t}(x) \leq 1$ and (A.4). For $x \geq 0$, (3.6) and (A.1) give $Q_{u,t}(x) \leq H(x) + e^{-2ux/t}$. The first term contributes at most $(\beta/a)e^{\kappa x}H(x)$, whereas direct Gaussian integration gives

$$e^{\kappa u} \int_0^\infty \beta e^{-\beta t} \phi_t(u + ct - x) e^{-2ux/t} dt = \frac{\beta}{a} e^{au+cx-a(u+x)} = \frac{\beta}{a} e^{-(a-c)x}.$$

Thus an envelope is

$$(A.9) \quad D(x) = \begin{cases} (\beta/a)e^{\kappa x}, & x < 0, \\ (\beta/a)\{e^{\kappa x}H(x) + e^{-(a-c)x}\}, & x \geq 0. \end{cases}$$

Lemma A.1, with any $r \in (\kappa, \gamma_*)$, proves integrability. Replacing $H(x)$ by its uniform bound $D_r e^{-rx}$ gives a common envelope for all environments.

For (A.7), restrict the endpoint integral to $|t - u/a| > \Lambda s_u$. For fixed x and $u > x$, its scaled value is at most

$$\frac{\beta}{a} e^{\kappa x} \nu_{u-x}\{|t - u/a| > \Lambda s_u\},$$

and is also bounded by $D(x)$. Normal convergence, reverse Fatou under D , and then dominated convergence as $\Lambda \rightarrow \infty$ prove the claim. $\square$

Lemma A.3. *Let*

$$E_R(u) = \{\exists t \geq 0 : |t - \tau| > R, Y(t) > u\}.$$

Then

$$(A.10) \quad \lim_{R \rightarrow \infty} \limsup_{u \rightarrow \infty} e^{\kappa u} \mathbb{P}(E_R(u)) = 0.$$

The convergence is uniform over all admissible environments.

Proof. For $x \in \mathbb{R}$ define

$$b_R(x) = \mathbb{P}\{\sup_{s \geq R} (W(s) - (a/2)s) > x\}, \quad d_R(x) = \mathbb{P}\{\sup_{s \geq R} (B(s) - cs) > x\},$$

where B is a standard Brownian motion. Both bounds decrease to zero as $R \rightarrow \infty$. Following one specified child at every split gives a standard Brownian lineage, so $\mathbb{P}\{\sup_{s \geq R} L(s) > x\} \leq d_R(x)$ for every environment. Under endpoint conditioning, a backward crossing at age $r > R$ is, by (3.7), a crossing of $W(v) - (u/t)v > x$ at $v = r/(1 - r/t) \geq r$. Thus when $u/t \geq a/2$, the conditional probability of $E_R(u)$ is at most $b_R(x) + d_R(x)$, for every real x .

Let $J_{u,R}(x)$ be the scaled endpoint integral for this event. As $E_R(u)$ is an ordinary crossing event, $0 \leq J_{u,R}(x) \leq D(x)$. For fixed x and $u > x$, split the time integral according as $u/t \geq a/2$ or $u/t < a/2$. Using the bound above on the first region and the bound 1 on the second, (A.5) and (A.8) give

$$J_{u,R}(x) \leq \frac{\beta}{a} e^{\kappa x} [b_R(x) + d_R(x) + \nu_{u-x}\{t : u/t < a/2\}].$$

The last probability tends to zero by Lemma A.2. Consequently

$$\limsup_{u \rightarrow \infty} J_{u,R}(x) \leq \min \left\{ D(x), \frac{\beta}{a} e^{\kappa x} (b_R(x) + d_R(x)) \right\}.$$

Reverse Fatou under D , followed by dominated convergence as $R \rightarrow \infty$, proves (A.10). Retaining D is essential; the backward bound with drift $a/2$ alone need not be integrable against $e^{\kappa x}$. $\square$

Lemma A.4. *For fixed x and $\delta > 0$, under ν_{u-x} ,*

$$(A.11) \quad \left(\frac{t - u/a}{s_u}, t \bmod \delta \right) \implies (G, \Theta),$$

where G is standard normal and Θ is uniform on $[0, \delta)$, independently of G .

Proof. By Lemma A.2, replacing $f_{u,x}$ by the normal density ϕ changes bounded test expectations by $o(1)$. For a bounded continuous g and a phase Fourier mode of integer frequency j , the resulting expectation is

$$e^{2\pi i j u / (a\delta)} \int_{\mathbb{R}} g(y) \phi(y) e^{2\pi i j s_u y / \delta} dy.$$

It tends to zero for $j \neq 0$ by the Riemann–Lebesgue lemma, and equals $\mathbb{E}\{g(G)\}$ for $j = 0$. Uniform approximation by trigonometric polynomials and normal tightness prove joint convergence and independence. The phase may first be viewed on the circle; the uniform limit gives zero mass to its endpoint, so the assertion on $[0, \delta)$ follows. $\square$

Proof of Remark 2.8. Fix $\beta = 1$, $c = 3$, $k_i \in \{2, 3\}$ and $A_i = I_{k_i}$ throughout. Write $q_{\mathcal{E},h} = p_{\text{br},h}$ for the prescribed environment and $q_{k,h}$ for homogeneous offspring $k \in \{2, 3\}$, and put $\lambda_k = 3 + \sqrt{9 - 2(k-1)}$. In the homogeneous k -ary model, the nonnegative mean-one martingale

$$\sum_{v \in N(t)} e^{\lambda_k (X_v(t) - 3t)}$$

and its maximal inequality give $q_{k,h}(x) \leq q_{k,0}(x) \leq e^{-\lambda_k x}$ for $x > 0$; the same bound is trivial for $x \leq 0$.

Let σ_n be split n , with $\sigma_0 = 0$, and recall $n_j = 1 + \sum_{i=1}^j (k_i - 1)$. If $k_i = k$ for all $i > n$, cutting at σ_n leaves n_n conditionally independent homogeneous families. Any successful window starting before the cut entails a crossing in the pre-cut tree. Conditional on its lifetime tree, the n_n terminal ancestral paths are marginally Brownian, so the probability of such a crossing is at most $n_n e^{-6u}$. Apply (2.13) to the future families and dominate their scaled union probability by $\sum_{v \in N(\sigma_n)} e^{\lambda_k (X_v(\sigma_n) - 3\sigma_n)}$. Gaussian conditioning and the independent holding times of rates n_j give, for every fixed $h \geq 0$,

$$(A.12) \quad \begin{aligned} q_{\mathcal{E},h}(u) &\sim B_{k,h} H_{n,k} e^{-\lambda_k u}, \\ H_{n,k} &= n_n \mathbb{E} \left\{ e^{-(k-1)\sigma_n} \right\} = n_n \prod_{j=0}^{n-1} \frac{n_j}{n_j + k - 1} > 0. \end{aligned}$$

This expectation is finite and $\lambda_k < 6$, justifying both dominated convergence and removal of the pre-cut error. This passage uses only the integrable first-moment weight; no second-moment bound is needed.

Next, fixed-level probabilities are uniformly determined by a long enough initial offspring sequence. Indeed, $\mathbb{E}\{|N(t)|\} \leq e^{2t}$. Cover crossings in $[j, j+1]$ by the ancestral paths at $j+1$ and apply exponential Markov inequality with parameter 3 to each Brownian path. For some finite constant C , uniformly over the environment,

$$\mathbb{P}\{\exists t > T, v \in N(t) : X_v(t) - 3t > u\} \leq C e^{-3u - (5/2)\lfloor T \rfloor}.$$

Couple environments agreeing through split n to agree through σ_n . On $\{\sigma_n > T\}$, a difference between their Parisian events entails a crossing after T in at least one model. Consequently,

uniformly also in $h \geq 0$,

$$|q_{\mathcal{E},h}(u) - q_{\tilde{\mathcal{E}},h}(u)| \leq \mathbb{P}\{\sigma_n \leq T\} + 2Ce^{-3u-(5/2)\lfloor T \rfloor}.$$

Since $n_j \leq 1 + 2j$, the first term tends to zero uniformly as $n \rightarrow \infty$, by comparison with the nonexplosive ternary population. Choose T first and then n .

Construct the sequence inductively. At stage m , complete the already fixed finite prefix by constant offspring, alternating $k = 2$ and $k = 3$ between stages. By (A.12), choose $u_m > u_{m-1} + 1$ so that the provisional logarithmic rate is within $1/m$ of λ_k for each delay $h = 0, 1, \dots, m$. Fix a longer prefix so that every continuation changes each of these probabilities at u_m by relative error less than $1/(m+1)$; the preceding uniform estimate permits this choice. The resulting single sequence has rates tending to λ_2 and λ_3 along alternate subsequences, for every fixed integer delay. For real $h \geq 0$, choose an integer $H \geq h$ and use $q_{\mathcal{E},H} \leq q_{\mathcal{E},h} \leq q_{\mathcal{E},0}$. The same two subsequential limits follow by squeezing. $\square$

APPENDIX B. SPECTRAL INTERPRETATIONS

The constants also admit a stationary spectral interpretation. Put

$$(B.1) \quad F(t) = e^{\kappa Z(t)}, \quad I(f) = \int_{\mathbb{R}} f(s) ds, \quad b = \mathbb{E}\{I(F)\}.$$

The strengthened moment bound (B.7) below gives $0 < b < \infty$. The original F is not itself Brown–Resnick stationary: $\mathbb{E}\{F(-r)\} = e^{-\beta r}$ for $r > 0$, whereas $F(0) = 1$. Instead, choose a path and an origin (F, U) jointly under

$$(B.2) \quad \mathbb{P}_*(df, ds) = \frac{f(s)}{b} \mathbb{P}_F(df) ds, \quad \widehat{F}(t) = \frac{F(t+U)}{F(U)},$$

where $\mathbb{P}_F$ is the original law of F . Thus the path law is biased by $I(F)$ and, given the path, the origin has density $F(s)/I(F)$. For $t_1, \dots, t_n \in \mathbb{R}$ and $x_1, \dots, x_n > 0$, Tonelli's theorem gives the exponent function

$$\mathbb{E}_* \left\{ \max_{1 \leq i \leq n} \frac{\widehat{F}(t_i)}{x_i} \right\} = \frac{1}{b} \mathbb{E} \left\{ \int_{\mathbb{R}} \max_{1 \leq i \leq n} \frac{F(t_i + s)}{x_i} ds \right\}.$$

A change of integration variable proves translation invariance; one coordinate gives $\mathbb{E}_* \{\widehat{F}(t)\} =$

1. Thus $\widehat{F}(0) = 1$ and $\widehat{F}$ generates a stationary max-stable process with unit Fréchet margins. Equivalently, it satisfies the tilt–shift identity of [8, Theorem 2.5]. Its exponent measure has the mixed moving maxima representation

$$(B.3) \quad \nu_F(A) = \frac{1}{b} \int_{\mathbb{R}} \int_0^\infty \mathbb{P}\{rF(\cdot - s) \in A\} \frac{dr}{r^2} ds.$$

It is invariant under time translations and satisfies $\nu_F(rA) = r^{-1}\nu_F(A)$ and $\nu_F\{f : f(0) > 1\} = 1$.

For a positive continuous path f define

$$\begin{aligned} \mathcal{P}_{0,h}(f) &= \sup_{s \in \mathbb{R}} \inf_{0 \leq r \leq h} f(s+r), \\ \mathcal{P}_{\delta,m}(f) &= \frac{1}{\delta} \int_0^\delta \sup_{j \in \mathbb{Z}} \min_{0 \leq \ell \leq m} f((j+\ell)\delta - \theta) d\theta. \end{aligned}$$

Both functionals are invariant under time translations and positively homogeneous; the phase average ensures translation invariance in the grid case. For either observation rule $\star = (0, h)$ or (δ, m) , the constants have the alternative expressions

$$(B.4) \quad C_\star = \frac{\beta}{a\kappa} \mathbb{E} \{ \mathcal{P}_\star(F) \} = \frac{\beta b}{a\kappa} \mathbb{E}_\star \left\{ \frac{\mathcal{P}_\star(\widehat{F})}{I(\widehat{F})} \right\}.$$

Indeed, the ratio inside the last expectation equals $\mathcal{P}_\star(F)/I(F)$, and integration over the new origin cancels $I(F)$. The expectation in this representation is under $\mathbb{P}_\star$; normalizing F by its area under the original law would generally give a different constant. For continuous observation with $h = 0$, the expectation on the right of (B.4) is the Pickands constant of this dissipative max-stable process; see [9, Theorems 2.1 and 2.3]. More generally, the normalized continuous constant is the spectral Parisian Pickands rate

$$(B.5) \quad \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_\star \left\{ \sup_{0 \leq s \leq T} \inf_{0 \leq r \leq h} \widehat{F}(s+r) \right\} = \frac{a\kappa}{\beta b} C_{0,h}.$$

The short domination argument is given after (B.7).

A genealogical counterpart is available when $k_i \equiv k$ and $R_i \equiv R$. Label the splitting vertices by $\mathcal{T} = \{1, \dots, k\}^*$, with $\emptyset$ representing the first split, and measure time and position from that split. If T_v and X_v are the time and position of splitting vertex v , set

$$Q_v = \exp\{\kappa X_v - \kappa^2 T_v/2\}, \quad T_\emptyset = X_\emptyset = 0, \quad Q_\emptyset = 1.$$

For child j born at v , with lifetime τ_{vj} ,

$$Q_{vj} = Q_v M_{v,j}, \quad M_{v,j} = \exp\{\kappa W_j^{(v)}(\tau_{vj}) - \kappa^2 \tau_{vj}/2\},$$

where $\mathbf{W}^{(v)}$ is the sibling Brownian vector born at v . The offspring vectors $(M_{v,j})_{j=1}^k$ are independent and identically distributed over v , their coordinates may be dependent, and $\mathbb{E} \{M_{v,j} \mid \tau_{vj}\} = 1$. Consequently

$$(Q_{uv})_{v \in \mathcal{T}} \stackrel{d}{=} Q_u (Q_v^{(u)})_{v \in \mathcal{T}},$$

where $Q^{(u)}$ has the law of Q and is independent of Q_u , with $\mathbb{E} \{Q_u\} = 1$. This is the branch-regenerative construction of [10, Section 2.2]: Q is a branch-Brown–Resnick spectral cascade, and its associated max-stable field is invariant in law under the descendant shifts $v \mapsto uv$.

The link with our scalar cluster is explicit. If $\mathcal{C}(t)$ denotes the particles alive in the fresh cohort at relative time t , then $\kappa^2/2 - \kappa c = \beta$ gives

$$(B.6) \quad F(t) = e^{\beta t} \min_{v \in \mathcal{C}(t)} \exp\{\kappa X_v(t) - \kappa^2 t/2\}, \quad t \geq 0.$$

Thus the forward shape is a lower-envelope functional of the compensated genealogical field along the evolving population. Time translations in (B.3) and descendant shifts on $\mathcal{T}$ describe different invariance properties. The fresh-subtree factorization above uses $k_i \equiv k$ and $R_i \equiv R$ and need not hold for the prescribed chronological offspring and covariance sequences allowed in Theorem 1.1.

Envelope and Parisian rate. Take

$$0 < \varepsilon < \min\{c, \beta/\kappa, (a-c)/2\}.$$

Apply Lemma A.1 with drift $c - \varepsilon$ and (A.1) with drift $a - \varepsilon$. Since $\kappa^2/2 - \kappa(c - \varepsilon) = \beta + \kappa\varepsilon < 2\beta$ and $\kappa < 2(a - \varepsilon)$, they give

$$(B.7) \quad \begin{aligned} F(t) &\leq A_\varepsilon e^{-\kappa\varepsilon|t|}, & t \in \mathbb{R}, \\ A_\varepsilon &= \exp\{\kappa \sup_{s \in \mathbb{R}} (Z(s) + \varepsilon|s|)\}, & \mathbb{E}\{A_\varepsilon\} < \infty. \end{aligned}$$

This proves $0 < b < \infty$ and finiteness of ν_F on local-supremum exceedance sets.

To verify (B.5), write $q_h(s) = \inf_{0 \leq r \leq h} F(s + r)$. Tonelli's theorem and (B.2) yield

$$\frac{1}{T} \mathbb{E}_* \left\{ \sup_{0 \leq s \leq T} \inf_{0 \leq r \leq h} \widehat{F}(s + r) \right\} = \frac{1}{bT} \mathbb{E} \left\{ \int_{\mathbb{R}} \sup_{v \in [s, s+T]} q_h(v) \, ds \right\}.$$

For each path, the integral divided by T tends to $\sup_{\mathbb{R}} q_h$: this follows first for compactly supported functions and then by truncation using (B.7). Its value is bounded by $A_\varepsilon(1 + 2/(\kappa\varepsilon T))$, since $q_h \leq F$. Dominated convergence proves (B.5).

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